

Do Subjective Mortality Expectations Matter for Predicting Longevity?

Thelonious Goerz
Department of Sociology
PUBPOL 5390

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Twin Motivations: (1) Policy

- The population is aging, with some of the highest growth occurring for those over 65+
- Predicting longevity is important for population health research because it provides crucial input information to guide current and future decisions about how to care and support an aging population



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Konrad-Zuse-Strasse 1 · D-18057 Rostock · Germany · Tel +49 (0) 3 81 20 81 - 0 · Fax +49 (0) 3 81 20 81 - 202 · www.demogr.mpg.de

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The Limits of Predicting Individual-Level Longevity

Luca Badolato
Ari Decher-Frahn
Nicholas J. Ivers
Maria Miranda
Erin Walk
Elnura Zhaliyeva
Monica Alexander
Ugo Filippo Bassilini | bassilini@demogr.mpg.de
Emilio Zagheni | office-zagheni@demogr.mpg.de

Figure 1: Badolato et al. 2023

Twin Motivations: (2) Substantive

- ▶ Sociology ([Merton 1948](#); [Vaisey 2009](#)), and economics ([Velásquez-Giraldo, n.d.](#); [Bénabou and Tirole 2016](#)) recognize the theoretically predictive power of individuals' beliefs (e.g., financial decisions, health, etc.)
- ▶ Subjective beliefs about survival may matter for predicting longevity in prediction models

Background

- ▶ Past work shows that predicting longevity is challenging and highly uncertain ([Breen and Seltzer 2023](#); [Badolato et al. 2023](#))
- ▶ Moreover, correlations between individual mortality expectations and death age are weak, but their predictive power remains unclear
- ▶ There are also stark inequalities in predictability across groups (e.g., race, education, gender)
 - ▶ Subjective information may help

Present Study

- ▶ Idea: Lets bring these ideas to bear with a prediction exercise
- ▶ Goal: Predict longevity (death age) and survival past 75 years of age
- ▶ Evaluate the predictability using (1) Covariate feature set, (2) Covariates + Subjective Probabilities
- ▶ Conceptually, we think of the subjective probabilities as “utility” features – they enrich predictions made with other, more powerful, covariates

Data

- ▶ Health and Retirement Study
- ▶ Panel dataset of 32k individuals representative of US population 50+, (170,000 person-observations)
- ▶ Rich feature set including health, wealth, education, work, stocks, social network information, caregiving

Machine Learning Framework

Table 1: Summary of Machine Learning Models by Outcome

Outcome	Models	Evaluation
Longevity	OLS, LASSO, Ridge, Elastic Net, Tree, Random Forest, XGBoost	RMSE, R^2
Survival to 75	Logit, LASSO, Ridge, Elastic Net, Classification Tree, Random Forest, XGBoost	Accuracy, Precision, Sensitivity, AUC

Data Processing and Tuning Model Parameters

- ▶ For each model:
 - ▶ Reshape data to wide format
 - ▶ Filter to sample of those who have died (17.2k)
 - ▶ Impute missing predictors (mean,mode) and create missing indicators ([Mullainathan and Spiess 2017](#))
 - ▶ Recode all factor and character variables to one-hot indicators
- ▶ Use 5-fold cross-validation to tune parameters
 - ▶ All parameters are tuned except for:
 - ▶ Learn rate (XGboost)
 - ▶ N Trees (Tree, Random Forest, XGBoost)

Main Results: Absolute Improvement in Longevity

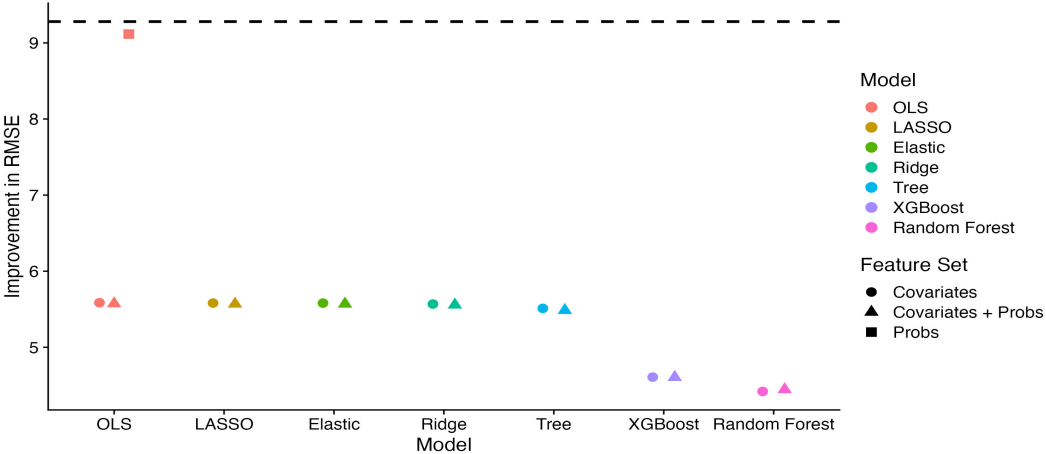


Figure 2: Absolute Improvement in Longevity Prediction

Main Results: Relative Improvement in Longevity

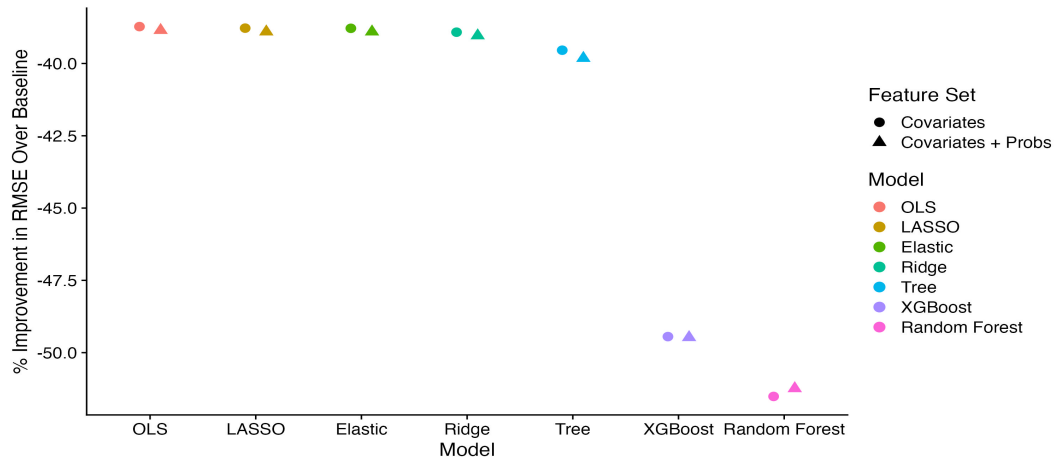


Figure 3: Relative Improvement in Longevity Prediction

Main Results: Relative Improvement in Longevity (Quintile)

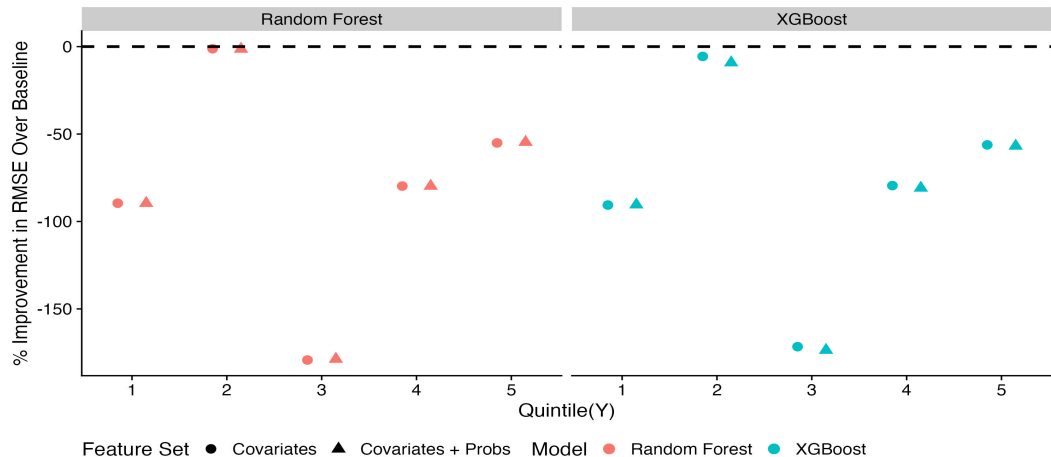


Figure 4: Relative Improvement in Longevity Prediction by Quintile of Death Age (Best Fitting Models)

Main Results: Absolute Improvement in Survival to 75

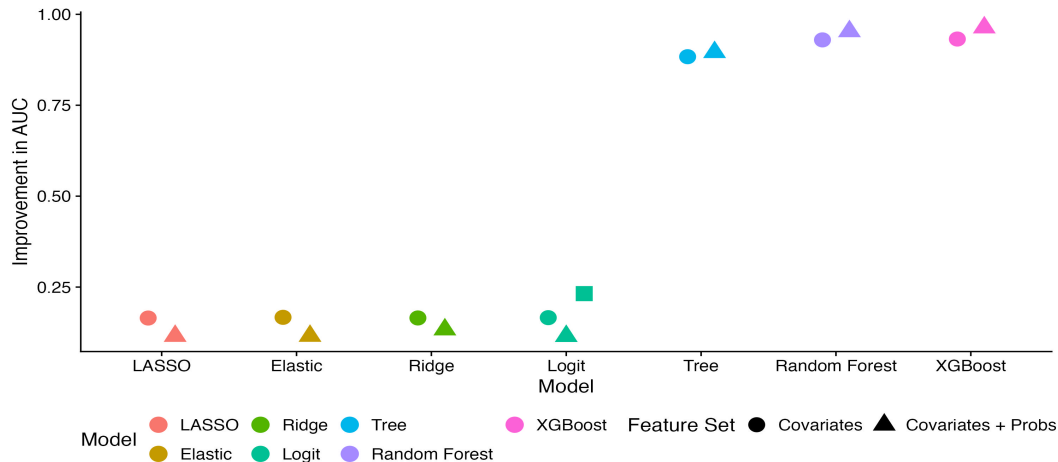


Figure 5: Absolute Improvement in Survival Threshold Prediction

Main Results: Relative Improvement in Survival to 75

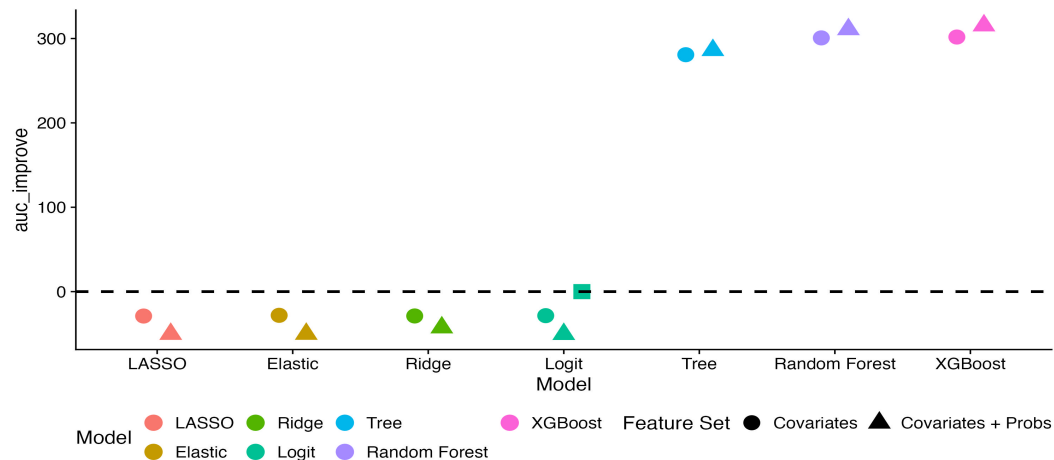


Figure 6: Relative Improvement in Survival Threshold Prediction

Main Results: Survival to 75 (Calibration Plot)

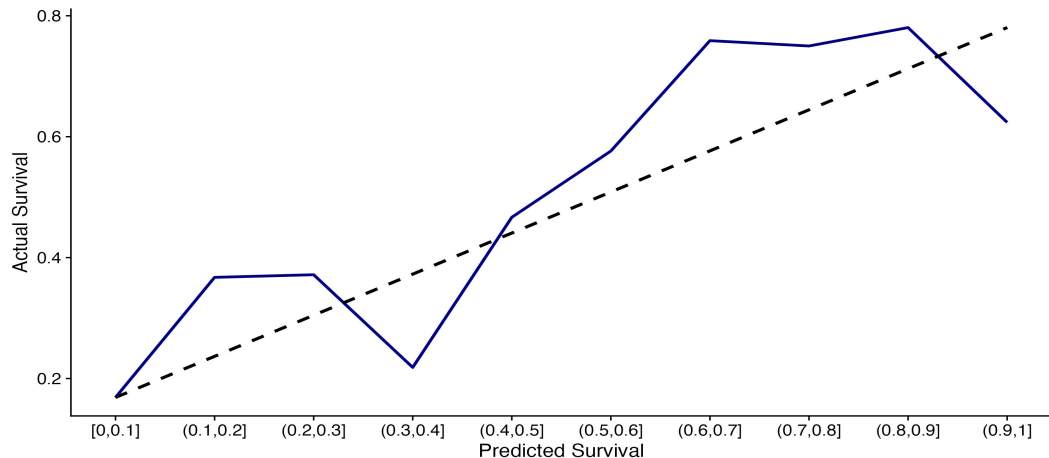


Figure 7: Calibration Plot of Full Feature Set XGBoost Algorithm Comparing Predicted Survival Probabilities and Actual Shares

Conclusion

- ▶ Tree-based methods with rich covariates improve on the baseline predictability of longevity and survival substantially (50% and 300% respectively)
- ▶ Across models, subjective mortality expectations do seem to aid in prediction, but only in small magnitudes and among certain groups
- ▶ Future efforts to collect data on subjective mortality could explore new ways of asking the questions (probabilities are hard for people to think about)

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